Adarsh Kumar Jha

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EXPERIENCE

AI Engineer (Production Engineering) Jan 2026 – Present (Intern to Full-Time, Jun 2026)
TBO.com (Travel Boutique Online)

- Architected near-real-time ETL/ELT lakehouse pipelines on S3 using Apache Spark, AWS Glue, Kafka CDC (Debezium), and Apache Hudi; enabled sub-second incremental processing across billions of records.
- Streamlined AWS Glue ETL pipeline reliability, elevating job success rate from 94% to 99.8% and cutting operating expenses by $300/day through resource tuning and schema caching.
- Restructured distributed data processing by routing ECS reads to same-AZ Kafka brokers, eliminating cross-AZ network latency and saving $5,000+/month in AWS transfer costs.
- Engineered a schema-aware Text-to-SQL ontology agent validating complex SQL on Amazon Athena, cutting BI team workload by 35%; exposed schema tools via MCP to Claude clients.
- Automated LLM classification of 650+ daily customer emails and 100,000+ KAM notes via AWS Lambda; applied prompt caching to slash inference spend by 70%+ and manual review by 57%.
- Pioneered a conversational AI voice bot (AWS Nova Sonic, RAG, tool calling) for automated IVR triage, sustaining p95 response latency under 600ms.

ML Engineer Intern May 2025 – Oct 2025
IIIT-Delhi – Research to Commercialization (R2C) | Advisor: Dr. Anuj Grover

- Built a commercialization matchmaking platform with FastAPI REST APIs and multi-agent AI document parsing, advancing project toward INR 40 Lakhs ($50K USD) NRDC funding.
- Calibrated document intelligence models in PyTorch with LoRA and PEFT, reaching 97% extraction accuracy, 75% parameter compression, and 2x faster inference.

Independent Research Project – Ontology Path Extraction Aug 2024 – Dec 2024
IIIT-Delhi x Oracle | Advisors: Prof. Raghava Mutharaju, Suman Roy (Oracle)

- Fine-tuned R-BERT in PyTorch with span-based embeddings, surpassing LLaMA-3 and GPT-3 baselines by 6 F1 points; indexed 25,000+ biomedical entities via FAISS, reducing false positives by 20%.

EDUCATION

Indraprastha Institute of Information Technology Delhi (IIIT-Delhi) 2022 – 2026
B.Tech, Computer Science and Artificial Intelligence CGPA: 8.2 / 10

SELECTED PROJECTS

VICAS Lab Portal & Resume Knowledge Graph [Code] Aug 2025 – Apr 2026

- Deployed an AI talent marketplace on React, FastAPI, Docker, and AWS EC2 across 6 microservices with OAuth2 and CI/CD; successfully onboarded 160+ registered students.
- Formulated a Gemini RAG pipeline with FAISS semantic retrieval and Neo4j knowledge graphs; trained a GCN for skill link prediction with cross-encoder re-ranking scoring 91.4% precision.

Customer Churn Prediction – Ensemble ML Benchmarking Academic Research Project

- Benchmarked XGBoost, LightGBM, and CatBoost on 50,000+ customer profiles, optimizing hyperparameters with Optuna to achieve 89.4% ROC-AUC (a 12% gain over baseline trees).
- Evaluated precision-recall curves to minimize false negatives, saving simulated revenue loss by 18% across test cohorts.

ML Serving Diagnostics & Observability Platform [Code] Jan 2025 – Apr 2025
Python, FastAPI, Docker, Kubernetes, TensorFlow, PyTorch, GCP

- Designed an automated diagnostics engine for Kubernetes ML workloads, isolating root causes of OOM faults and latency spikes; slashed mean time to resolution by 40% across 5 production services.
- Implemented a FastAPI telemetry dashboard streaming node health to Google Cloud Storage, cataloging automated remediation runbooks that prevented service downtime.

TECHNICAL SKILLS

Languages: Python, SQL (Complex Queries, Optimization), Java, C, C++, JavaScript
Data Systems: Apache Spark, Apache Kafka, Debezium (CDC), AWS Glue, Apache Hudi, Parquet, Amazon Athena, ETL/ELT, Data Lakehouse, DynamoDB, Neo4j, Big Data Processing
AI / Machine Learning: PyTorch, LLMs, RAG, AI Agents, Tool Calling, MCP, Text-to-SQL, FAISS Vector Search, Embeddings, GCN, LoRA, PEFT, XGBoost, LightGBM, CatBoost
Cloud & DevOps: AWS (Athena, Glue, S3, Lambda, ECS, DynamoDB, CloudWatch, EC2), GCP (BigQuery), FastAPI, Docker, Kubernetes, Linux, Git, CI/CD

ACHIEVEMENTS & LEADERSHIP

Ranked AIR 252 in GATE 2026 (Computer Science & Information Technology) 2026
JEE Mains 2022: All India Rank 4778 (99.5 percentile) among 872,970 candidates 2022
Codeforces (Specialist - Max Rating 1450+), LeetCode 300+ problems solved, GFG 2023 – 2026
Operations Head, Odyssey IIIT-D – Led 30-member logistics council for cultural festival hosting 30,000+ attendees 2025